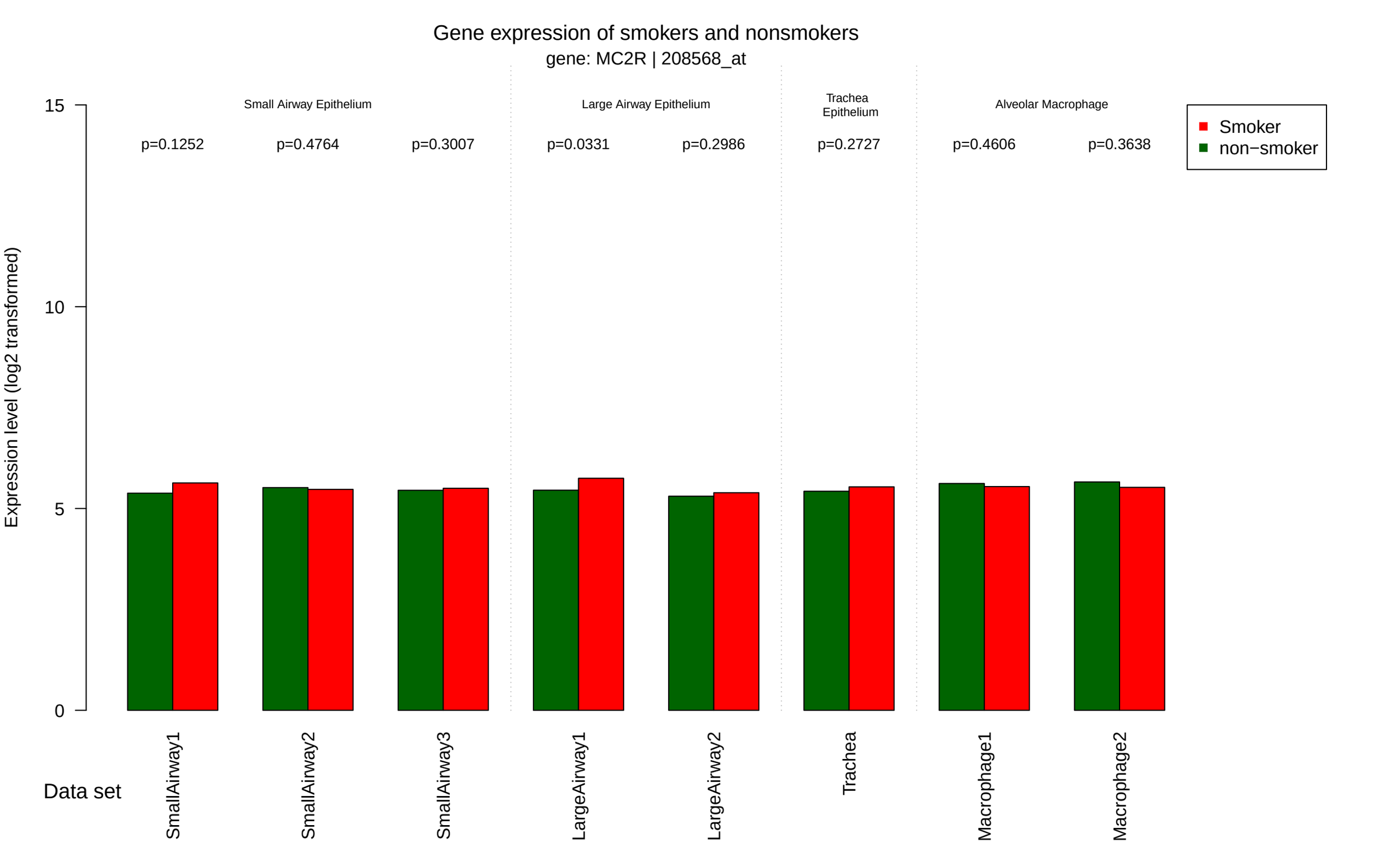
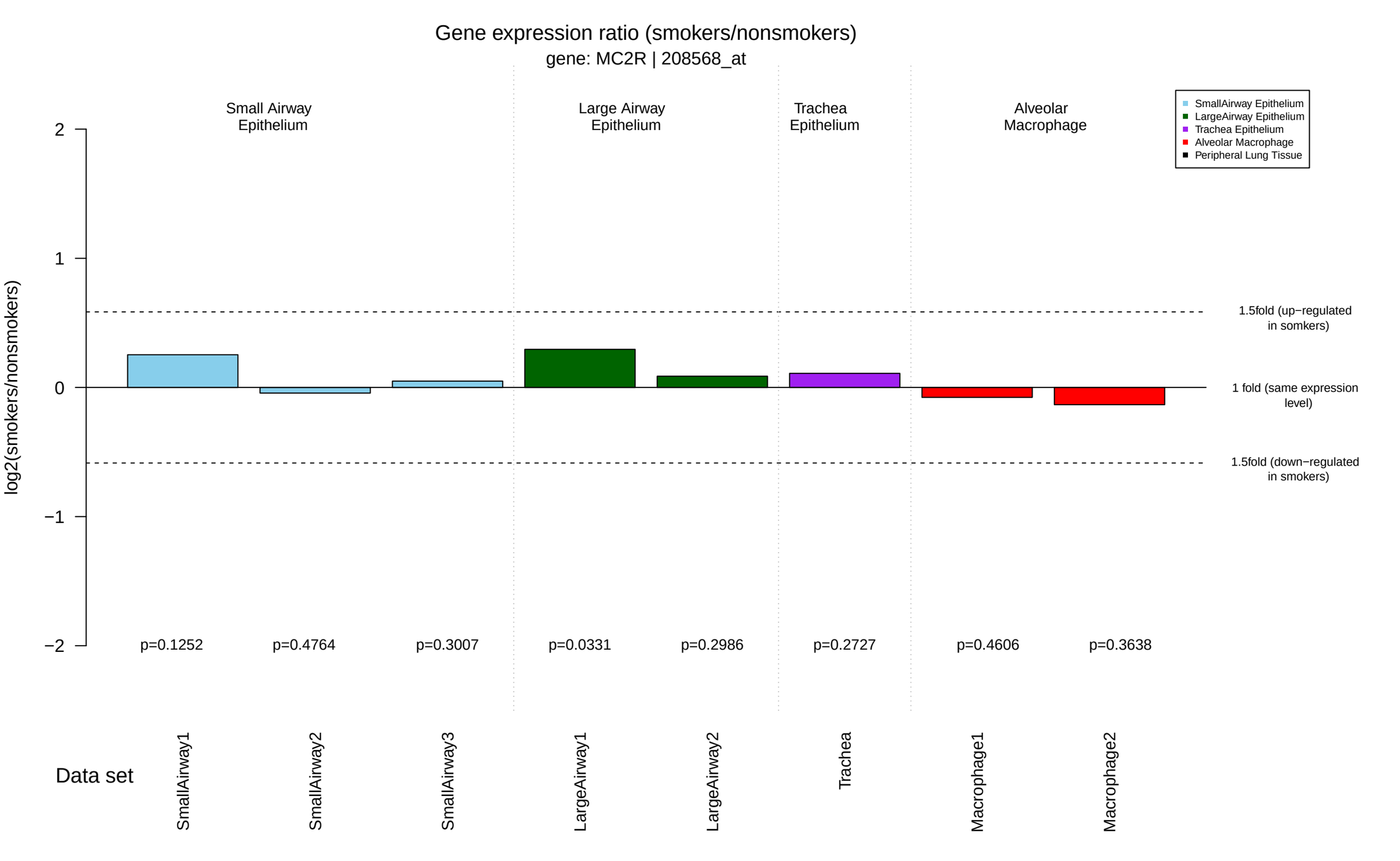
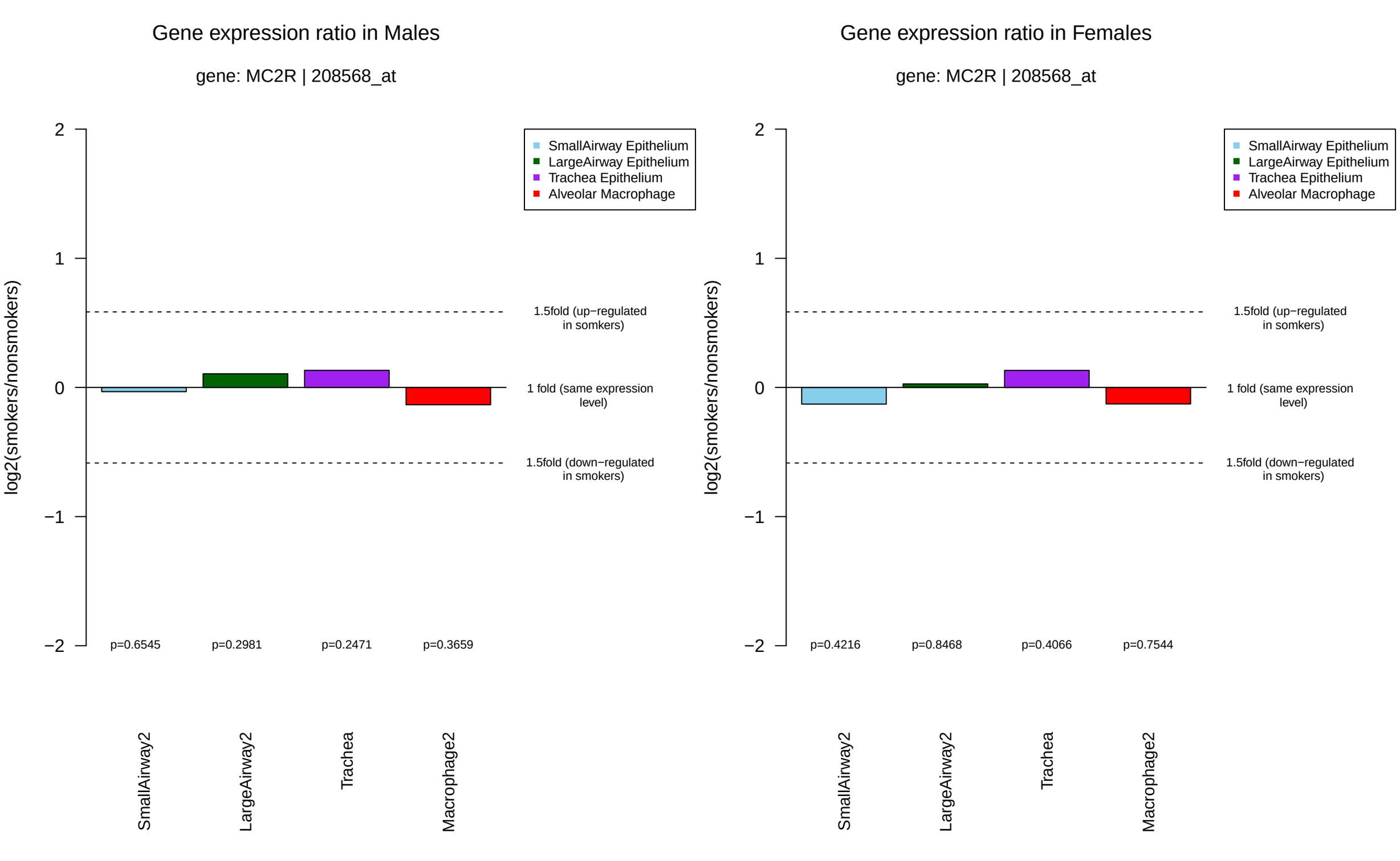
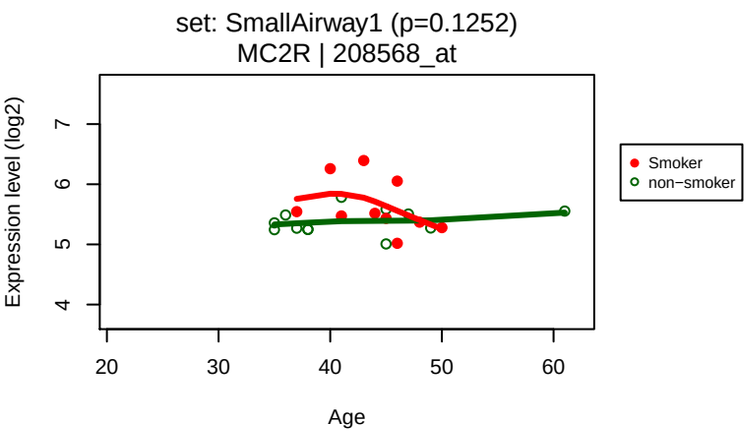
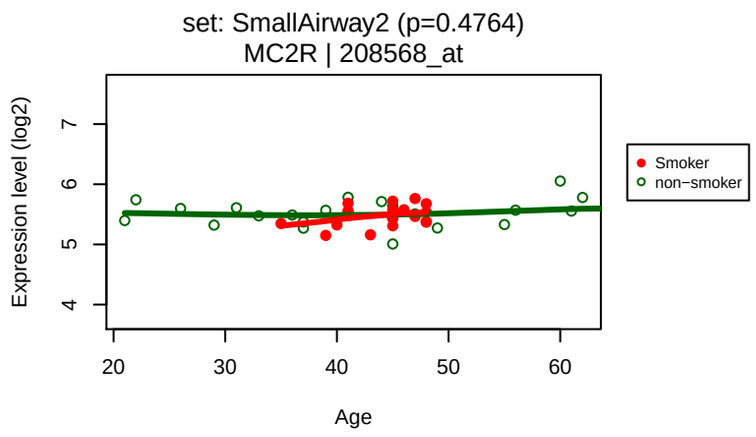
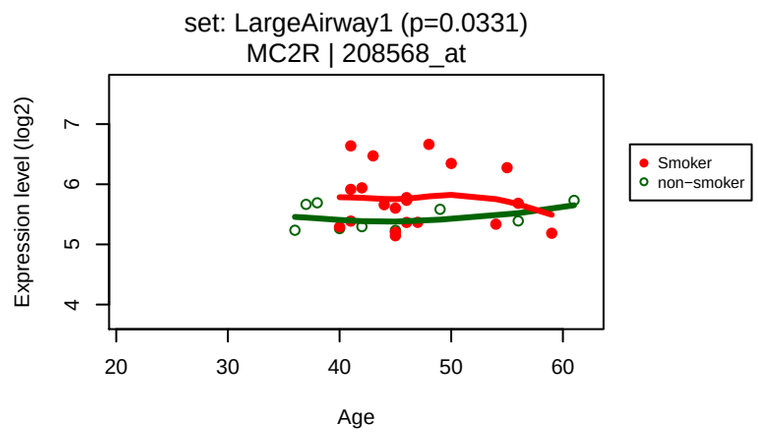

Gene expression of smokers and nonsmokers
gene: MC2R | 208568_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1252
p=0.4764
p=0.3007
p=0.0331
p=0.2986
p=0.2727
p=0.4606
p=0.3638
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

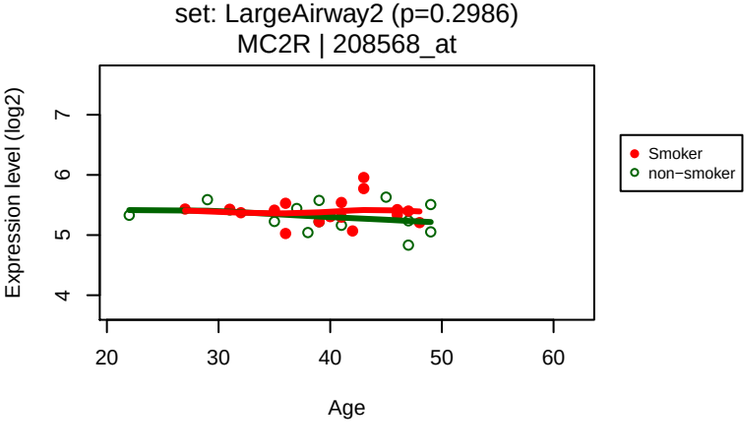

Gene expression ratio (smokers/nonsmokers)
gene: MC2R | 208568_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1252
p=0.4764
p=0.3007
p=0.0331
p=0.2986
p=0.2727
p=0.4606
p=0.3638
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

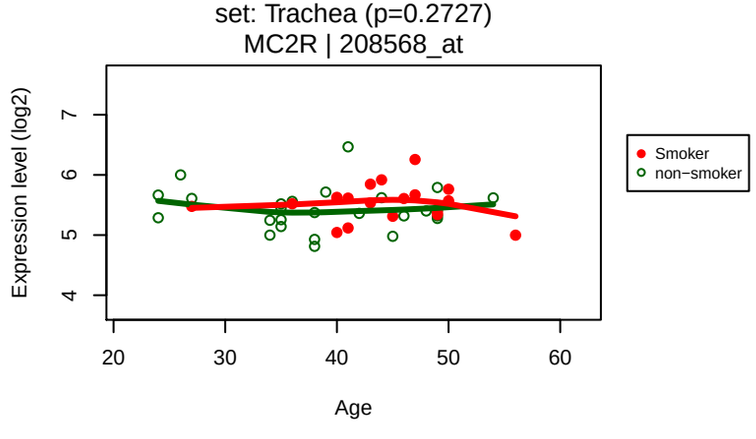

Gene expression ratio in Males
gene: MC2R | 208568_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6545
p=0.2981
p=0.2471
p=0.3659
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MC2R | 208568_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4216
p=0.8468
p=0.4066
p=0.7544
SmallAirway2
LargeAirway2
Trachea
Macrophage2

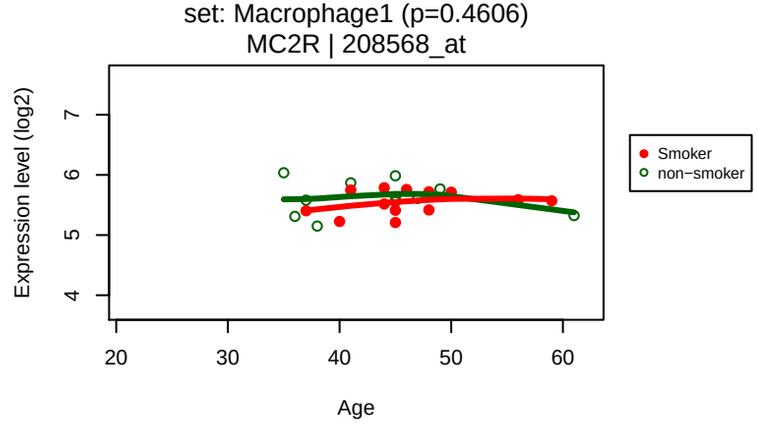

set: SmallAirway1 (p=0.1252)
MC2R | 208568_at
Expression level (log2)
Age
Smoker
non−smoker

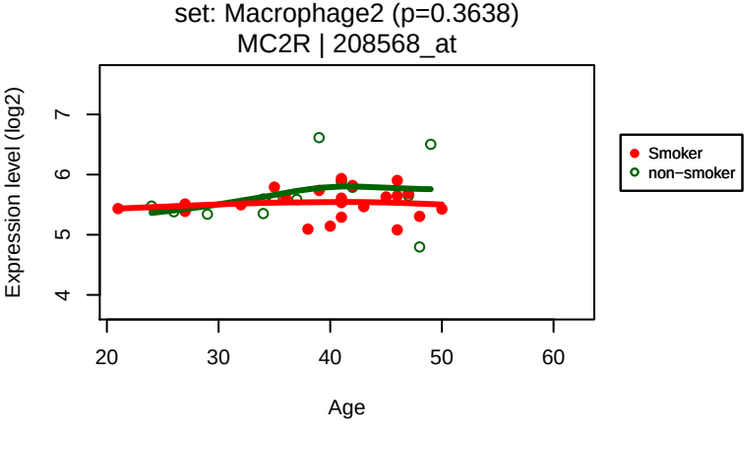

set: SmallAirway2 (p=0.4764)
MC2R | 208568_at
Expression level (log2)
Age
Smoker
non−smoker

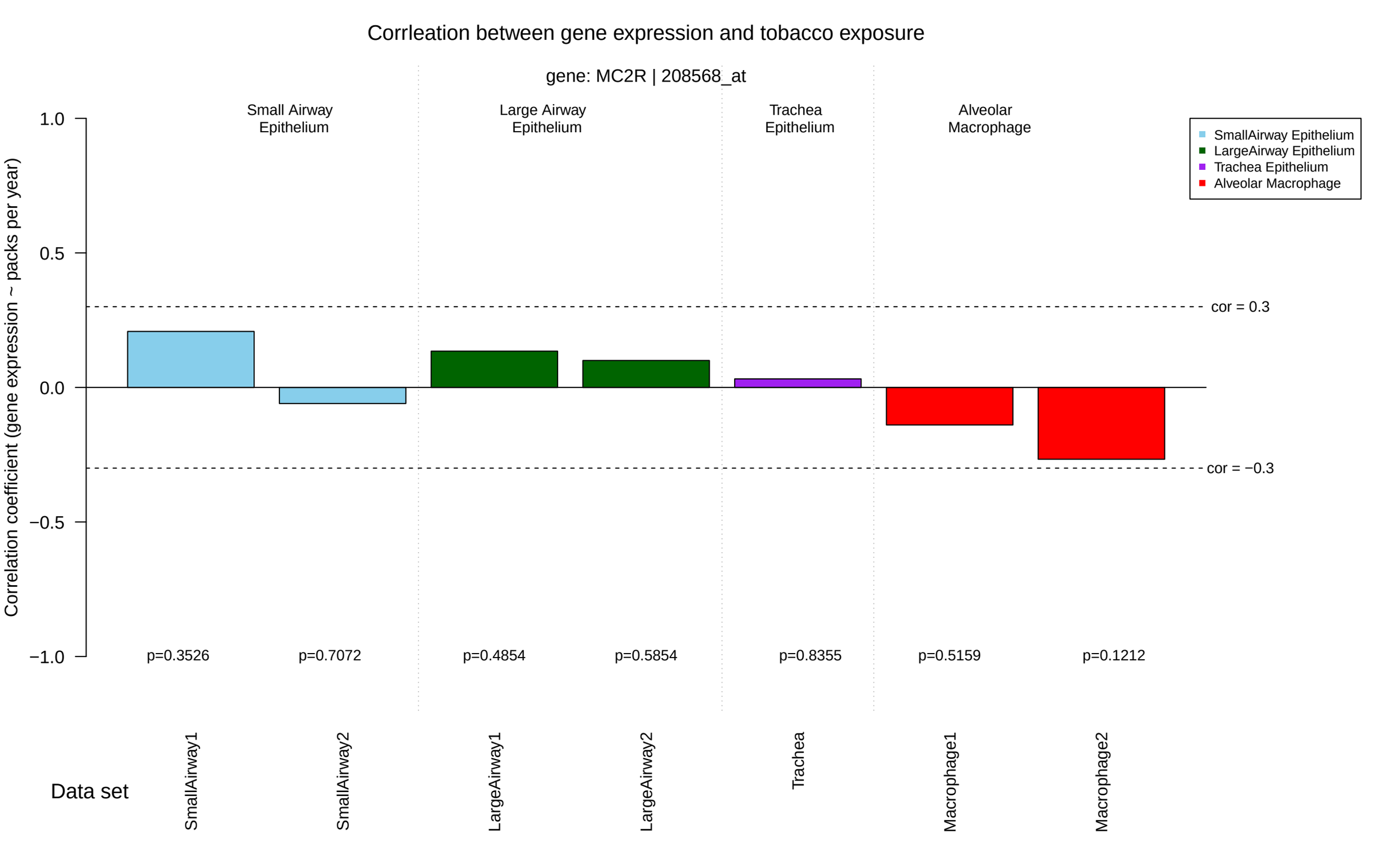

set: LargeAirway1 (p=0.0331)
MC2R | 208568_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2986)
MC2R | 208568_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2727)
MC2R | 208568_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4606)
MC2R | 208568_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3638)
MC2R | 208568_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MC2R | 208568_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3526
p=0.7072
p=0.4854
p=0.5854
p=0.8355
p=0.5159
p=0.1212
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2